Introduction

In Chapter 3, we learned how to fine-tune a model for a specific task. When we do that, we use the same tokenizer that the model used during pretraining. But what should we do if we want to train a model from scratch?

In that case, using a tokenizer trained on a different language or domain is usually not a good choice. For example, a tokenizer trained on English text will not work well on Japanese text, because the way spaces and punctuation are used is very different.

In this chapter, you will learn how to train a completely new tokenizer on a text dataset. Then that tokenizer can be used to pretrain a language model. We will do this using the Tokenizers library, which is used to make the fast tokenizers in the Transformers library.

We will look closely at the features of this library and also see how fast tokenizers are different from slow tokenizers.

Topics covered in this chapter

- How to train a new tokenizer like the one used by a checkpoint, but on a new text dataset
- The special features of fast tokenizers
- The differences between the three main subword tokenization methods used in NLP today
- How to build a tokenizer from the beginning with the Tokenizers library and train it on data

The ideas in this chapter will help you prepare for Chapter 7, where we will create a language model for Python source code.